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DIFFERENCE-IN-DIFFERENCES WITH VARIATION IN TREATMENT TIMING

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In contrast to our substantial understanding of the canonical 2x2 DD model, we know relatively little about the two-way fixed effects DD model when treatment timing varies. We do not know precisely how it compares mean outcomes across groups.⁵ We typically rely on general descriptions of the identifying assumption like “interventions must be as good as random, conditional on time and group fixed effects” (Bertrand, Duflo, and Mullainathan 2004, p. 250), and consequently lack well-defined strategies to test the validity of the DD design with timing. We have limited understanding of the treatment effect parameter that regression DD identifies. Finally, we often cannot evaluate when alternative specifications will work or why they change estimates.⁶

This paper shows that the two-way fixed effects DD estimator in (2) is a weighted average of all possible 2x2 DD estimators that compare timing groups to each other. Some use units treated at a particular time as the treatment group and untreated units as the control group. Some compare units treated at two different times, using the later group as a control before its treatment begins and then the earlier group as a control after its treatment begins. As in any least squares estimator, the weights on the 2x2 DD’s are proportional to group sizes *and* the variance of the treatment dummy within each pair. Treatment variance is highest for groups treated in the middle of the panel and lowest for groups treated at the extremes. This result clarifies the theoretical interpretation and identifying assumptions of the general DD model and creates simple new tools for describing the design and analyzing problems that arise in practice.

By decomposing the DD estimator into its sources of variation (the 2x2 DD’s) and providing an explicit interpretation of the weights in terms of treatment variances, my results

⁵ Imai, Kim, and Wang (2018) note “It is well known that the standard DiD estimator is numerically equivalent to the linear two-way fixed effects regression estimator if there are two time periods and the treatment is administered to some units only in the second time period. Unfortunately, this equivalence result does not generalize to the multi-period DiD design...Nevertheless, researchers often motivate the use of the two-way fixed effects estimator by referring to the DiD design (e.g., Angrist and Pischke, 2009).”

⁶ This often leads to sharp disagreements. See Neumark, Salas, and Wascher (2014) on unit-specific linear trends, Lee and Solon (2011) on weighting and outcome transformations, and Shore-Sheppard (2009) on age time fixed effects.

The challenge in this setting has been to articulate how estimates of equation (2) compare the groups and times depicted in figure 1. We do, however, have clear intuition, for 2x2 designs in which one group's treatment status changes and another's does not. We could form several such designs, estimable by equation (1), in the three-group case. Figure 2 plots them.

Panels A and B show that with only one of the two treatment groups, an estimate from equation (2) reduces to the canonical case comparing a treated to an untreated group:

$$\hat{\beta}_{jU}^{2x2} \equiv \left(\bar{y}_j^{POST(j)} - \bar{y}_j^{PRE(j)} \right) - \left(\bar{y}_U^{POST(j)} - \bar{y}_U^{PRE(j)} \right), \quad j = k, \ell. \quad (3)$$

If instead there were no untreated units, the two way fixed effects estimator would be identified only by the differential treatment timing between groups k and ℓ . For this case, panels C and D plot two clear 2x2 DD's based on sub-periods when only one group's treatment status changes. Before t_ℓ^* , the early units act as the treatment group because their treatment status changes, and later units act as controls during their pre-period. We compare outcomes between the window when treatment status varies, $MID(k, \ell)$, and group k 's pre-period, $PRE(k)$:

$$\hat{\beta}_{k\ell}^{2x2,k} \equiv \left(\bar{y}_k^{MID(k,\ell)} - \bar{y}_k^{PRE(k)} \right) - \left(\bar{y}_\ell^{MID(k,\ell)} - \bar{y}_\ell^{PRE(k)} \right) \quad (4)$$

The opposite situation, shown in panel D, arises after t_k^* when the later group changes treatment status but the early group does not. Later units act as the treatment group, early units act as controls, and we compare average outcomes between the periods $POST(\ell)$ and $MID(k, \ell)$:

$$\hat{\beta}_{k\ell}^{2x2,\ell} \equiv \left(\bar{y}_\ell^{POST(\ell)} - \bar{y}_\ell^{MID(k,\ell)} \right) - \left(\bar{y}_k^{POST(\ell)} - \bar{y}_k^{MID(k,\ell)} \right) \quad (5)$$

The already-treated units in group k can serve as controls even though they are treated because treatment status does not change.

My central result is that any two-way fixed effects DD estimator is a weighted average of well-understood 2x2 DD estimators, like those plotted in figure 2. To see why, first assume a

$\bar{D}_K$ is small. The opposite conclusions follow from “reverse Roy” selection where units with the smallest effects select into treatment first, which describes the take up of housing vouchers (Chyn forthcoming) and charter school applications (Walters forthcoming).

An easy way to gauge whether VWATT and a sample-weighted ATT is to scatter the weights from (13) against each group’s sample share. These two may be close if there is little variation in treatment timing, if the untreated group is very large, or if some timing groups are very large. Conversely, weighting matters less if the ATT_k ’s are similar, which one can evaluate by aggregating each group’s 2x2 DD estimates from the decomposition theorem.¹⁹ Finally, one could directly compare VWATT to point estimates of a particular parameter of interest. Several alternative estimators can deliver differently weighted averages of ATT’s (Abraham and Sun 2018, Callaway and Sant’Anna 2018, de Chaisemartin and D’Haultfœuille 2018b).

ii. Effects that vary over time but not across units

Time-varying treatment effects (even if they are identical across units) generate cross-group heterogeneity in VWATT by averaging time-varying effects over different post-treatment windows, but more importantly they mean that $\Delta ATT \neq 0$. Equations (11b) and (11c) show that common trends in counterfactual outcomes leaves one set of timing terms biased ($\hat{\beta}_{k\ell}^{2x2,\ell}$), while common trends between counterfactual and treated outcomes leaves the other set biased ($\hat{\beta}_{k\ell}^{2x2,k}$).

To illustrate this point, figure 3 plots a case where counterfactual outcomes are identical, but the treatment effect is a linear trend-break, $Y_{it}^1 = Y_{it}^0 + \phi \cdot (t - t_i^* + 1)$ (see Meer and West 2013). $\hat{\beta}_{k\ell}^{2x2,k}$ uses group ℓ as a control group during its pre-period and identifies the ATT during

¹⁹ Relatedly, de Chaisemartin and D’Haultfœuille (2018b) derive a statistic that gives the minimum variation in treatment effects that could lead to wrong-signed regression estimates when all treatment effects have the same sign.